

Arterio Rodrigues

rodrigues.arterio@gmail.com | [github](#) | [linkedin](#) | 646-626-2211

Education

Hunter College *MA in Computer Science*

January 2024 – May 2027

Hunter College *BA in Computer Science, Minor in Math*

January 2019 – May 2023

Google Software Engineer Program *Project Basta*

September 2023 – June 2024

AWS Certified Developer Associate

June 2024 – June 2027

Work Experience

JetBlue, *Associate Full Stack Engineer*

December 2024 – Present

- Designed and shipped a mobile application used by 20+ crewmembers at JFK terminal to manage lounge access, unlocking a new revenue stream for the business.
- Designed Azure SQL schemas for users, visitors, and runtime configuration; built NestJS backend services bridging the Next.js frontend to those databases and to third-party payment providers.
- Built the payment-processing flow end-to-end in TypeScript — transaction scanning, authentication, and audit logging — supporting the lounge app's revenue pipeline.
- Patched bugs on the legacy Angular jetblue.com site and migrated sections into a new Next.js codebase; co-authored a shared component toolkit porting legacy UI elements into Next.js.

Hunter College Career Center, *IT Specialist*

April 2022 – December 2024

- Wrote a Python pipeline that cut CSV data-integration time for 5,000 records from 2 weeks to 1 day, and added an automated file/folder sorter that routes uploads by approved file type.
- Built a full-stack web scraper in Node.js (HTML/CSS/JS) that crawled 3,000 links on the company site and surfaced 54 broken ones, replacing weeks of manual auditing with a same-day report.
- Authored Power BI dashboards over complex Excel datasets, using DAX measures to surface trends for stakeholders and drive a 20% improvement in reporting performance.

Projects

AdamOS—*32-bit x86 Operating System*

- Built a 32-bit OS from scratch with a custom bootloader that switches into protected mode, sets up the GDT/TSS, and hands control to a kernel running at ring 0.
- Implemented a preemptive PIT-driven scheduler, per-process page tables with 4KB paging, and full ring 0/ring 3 isolation via `int 0x80` syscalls including `fork`, `exec`, and `waitpid`.
- Wrote a FAT16 filesystem on a virtual ATA PIO disk and a user-space shell with isolated programs (vim-like editor, BF interpreter) running in their own address spaces.

React Rebuild—*Custom React Implementation*

- Reimplemented core React from scratch, including a virtual DOM, diffing-based reconciliation, and a functional component model.
- Built a hooks system (`useState`, `useEffect`-style) backed by per-component state slots and a render scheduler that re-renders only on state change.
- Designed two variants (class-style and functional) to compare rendering and state-management approaches.

Creature Evolution—*Genome-Encoded Neural Network Simulator*

- Built an evolution simulator where each creature carries a hex-string genome that decodes into a feed-forward neural network — sensor → internal → action neurons — driving behavior every tick.
- Implemented the genome decoder, connection-pruning pass (iteratively removes dead-end internals until stable), and the recurrent feed-forward pass with tanh activations and last-step internal state.
- Set up the SFML render loop, configurable simulation parameters (population, mutation rate, genome length), and a CMake build that auto-fetches SFML.

Arc Drone—*Custom Quadcopter Flight Controller*

- Designed and flashed firmware for a quadcopter, integrating a 6-axis IMU, four ESCs, an OLED telemetry display, and a wireless controller link.
- Built modular Arduino libraries for each peripheral (IMU, ESC, OLED, controller) so the main flight loop stays clean and each subsystem is independently testable.
- Implemented sensor reads, motor PWM output, and the controller-input pipeline that feeds the stabilization loop on the main `drone.ino` sketch.

Skills

Databases: PostgreSQL, MongoDB

Frameworks: React, React Native, Next.js, Angular, NestJS Power Bi

Programming Languages: C++, C, Python, TypeScript, JavaScript, MIPS Assembly, x86 Assembly

Cloud & DevOps: Azure (SQL Database), AWS (EC2, S3, Lambda, RDS), Docker, GitHub Actions, Linux